FLE TRIBE

BONUS COURSE · TYPESCRIPT · THURSDAY · LESSON GUIDE

Config & strict mode

This course: Add types to your JavaScript, the way most professional React teams work.

tsconfig.json

Every TypeScript project has a `tsconfig.json` that tells the compiler how strict to be and what JavaScript to produce. Vite creates one for you; you'll mostly change a few options.

```
{
  "compilerOptions": {
    "target": "ES2020",
    "strict": true,
    "noUnusedLocals": true,
    "noUncheckedIndexedAccess": true
  },
  "include": ["src"]
}
```

Turn strict mode on

`"strict": true` switches on a group of checks. The most important is **strictNullChecks**: TypeScript makes you handle values that might be `null` or `undefined` before you use them, which removes a whole class of "cannot read property of undefined" crashes.

```
function greet(name: string | null) {
  // return name.toUpperCase();  ✗ error: name might be null
  return name ? name.toUpperCase() : 'Guest'; // ✓
}
```

Typing API responses

Data from an API is `unknown` until you describe it. Write an interface for the shape you expect:

```
interface Weather {
  current: { temperature_2m: number; wind_speed_10m: number };
}

const res = await fetch(url);
const data: Weather = await res.json();
console.log(data.current.temperature_2m);
```

Reading error messages

- Read the **last line** first: it usually says what is wrong.
- Note the file and line number, then go there.
- "Type 'X' is not assignable to type 'Y'" means you gave the wrong kind of value.
- "Object is possibly 'undefined'" means you need to check before using it.

Research questions

Answer these in the app, in your own words, to unlock the next lesson.

1. What does `strictNullChecks` protect you from?
2. Type the Open-Meteo response you used in Week 5.

Start your research here

- TS Playground: https://www.typescriptlang.org/play